

Exercise 4

UNION AND UNION ALL

```
Q1 SELECT account_id,  
          account_holder,  
          'Sandton' AS city  
FROM branch_sandton_accounts  
UNION  
SELECT account_id,  
          account_holder,  
          'Rosebank' AS city  
FROM branch_rosebank_accounts  
ORDER BY account_holder;
```

Results

account_id	account_holder	city
Acc1	Mamula Dlamini	Sandton
Acc2	David Motswagole	Sandton
Acc3	Lerato Sithole	Sandton
Acc4	Peter Mlengi	Sandton

Q2 SELECT product_code,
 product_name,
 'Savings' AS product_type
 FROM savings_products
 UNION
 SELECT product_code,
 product_name
 'Current' AS product_type
 FROM current_products
 ORDER BY product_code;

Results

product_code	product_name	product_type
SU01	Basic Savings	Savings
SU02	Premium Savings	Savings
SU03	Youth Savings	Savings
SU04	Business Savings	Savings
CR01	Standard Current	Savings Current
CR02	Gold Current	Current
CR03	Business Current	Current

Q3 SELECT staff_id,
 staff_name,
 'email' AS 'Null' AS email
 FROM retail_banking_staff
 UNION
 SELECT staff_id,
 staff_name,
 'Null' AS email
 FROM corporate_banking_staff;

```

04 SELECT city-code,
           city-name,
           'mobile' AS region
FROM mobile-branch-cities
UNION
SELECT city-code,
           city-name,
           'digital' AS region
FROM digital-branch-cities
ORDER BY city-code;

```

Results

city-code	city-name	region
C01	Johannesburg	branch
C02	Pretoria	branch
C03	Cape Town	branch
C04	Durban	branch
C05	Potokwane	digital
C06	Port Elizabeth	digital

~~05 SELECT customer-id, customer-name, notification~~

```

05 SELECT customer-id,
           customer-name,
           'Push-notification' AS segment
FROM push-notification-targets
UNION
SELECT customer-id,
           customer-name,
           'In-app-banner' AS segment
FROM in-app-banner-targets
ORDER BY customer-id;

```

Results

customer_id	customer_name	Segment
C1001	Momona Zwane	Push-notification
C1002	Andile Buthelez	Push-notification
C1003	Fatima Mahomed	Push-notification
C1004	Ryno Van Zyl	Push-notification
C1005	Thandeka Cele	Inapp-banner
C1006	Samuel Mkesi	Inapp-banner

UNION ALL section

```
a SELECT transaction_id,  
      account_id,  
      amount,  
      CurrentDate() AS transaction_date  
FROM atm01-transactions  
UNION ALL  
SELECT transaction_id,  
      account_id,  
      amount,  
      CurrentDate() AS transaction_date  
FROM atm02-transactions  
ORDER BY transaction_id;
```

Results

transaction_id	account_id	amount	transaction_date
T1001	A001	500	15-07-2026
T1002	A002	1200	15-07-2026
T1003	A001	300	15-7-2026
T1004	A003	750	15-7-2026
T1003	A001	300	15-7-2026
T1005	A004	900	15-7-2026

T1006	A002	450	15-7-2026
T1007	A005	150	15-7-2026

app_id AS
 62 SELECT ↑ application_id,
 customer_id,
 loan_type,
 amount-requested
 FROM gauteng-loan-applications
 UNION ALL

SELECT app_id AS application_id,
 customer_id,
 loan_type,
 amount-requested

FROM western-cape-loan-applications;

Results

application-id	customer-id	loan-type	amount-requested
LA001	C1001	Home loan	800 000
LA002	C1002	Vehicle loan	350 000
LA003	C1003	Personal loan	150 000
LA004	C1004	Home loan	750 000
LA003	C1003	Personal loan	200 000
LA005	C1005	Vehicle loan	250 000
LA006	C1006	Business	500 000
LA007	C1007	Home loan	600 000

```

03 SELECT complaint_id,
    customer_id,
    'email' AS category,
    Current date ( ) AS logged_date
FROM email-complaints
UNION ALL
SELECT complaint_id,
    customer_id,
    'app' AS category,
    Current date ( ) AS logged_date
FROM app-complaints;

```

Results

complaint_id	customer_id	category	logged_date
EC001	C2001	email	15-07-2026
EC002	C2002	email	15-07-2026
EC003	C2003	email	15-07-2026
EC004	C2004	email	15-07-2026
AC001	C2005	app	15-07-2026
AC002	C2001	app	15-07-2026
AC003	C2006	app	15-07-2026
AC004	C2007	app	

```

04 SELECT payment_id,
    account_id,
    amount,
    'Null' AS payment_date
FROM april-payments
UNION ALL
SELECT payment_id,
    account_id,

```

amounts
 ('Null') AS payment-date
 FROM my-payments;

Results

payment-id	account-id	amount	payment-date
PAY001	Acc1	125	Null
PAY002	Acc2	420	Null
PAY003	Acc3	920	Null
PAY004	Acc4	330	Null
PAY005	Acc1	120000	Null
PAY006	Acc5	760000	Null
PAY007	Acc2	510000	Null
PAY008	Acc6	280000	Null

```

AS SELECT entry-id,
          account-id,
          entry-type,
          ('Null') AS amount,
          ('Null') AS entry-date
FROM debit-entries
UNION ALL
SELECT entry-id,
          account-id,
          entry-type,
          ('Null') AS amount,
          ('Null') AS entry-date
FROM credit-entries;
```


Results

entry-id	account-id	entry-type	amount	entry-date
DRC01	Acc1	Debit	Null	Null
DRC02	Acc2	Debit	Null	Null
DRC03	Acc3	Debit	Null	Null
DRC04	Acc4	Debit	Null	Null
CR001	Acc1	Credit	Null	Null
CR002	Acc5	Credit	Null	Null
CR003	Acc2	Credit	Null	Null
CR004	Acc6	Credit	Null	Null

Bonus Questions

1. UNION - The fraud analyst will be able to get a list of unique customers, thus if we opt for UNION ALL it would take the fraud analyst more time or causing them to investigate one customer multiple times since there will be duplicates.
2. UNION ALL - This would give the auditing team a full list including duplicates transactions done by individuals whereas if we opt for UNION it would take out duplicates leaving the auditing team with unique list of individuals.
- 3 The query fails because the number of columns does not match. The first Syntax only contains 2 columns whereas the second Syntax has 3 columns. The rule for UNION and UNION ALL states that there should be same number of columns and compatible data types should be in the same order.

New Syntax

```
SELECT account_id,  
       account_holder,  
       'Null' AS city  
FROM branch_sutton_accounts  
UNION
```

```
SELECT account_id,  
       account_holder,  
       city  
FROM branch_rosebank_accounts;
```

Another solution is to drop the 3rd column from the second syntax.